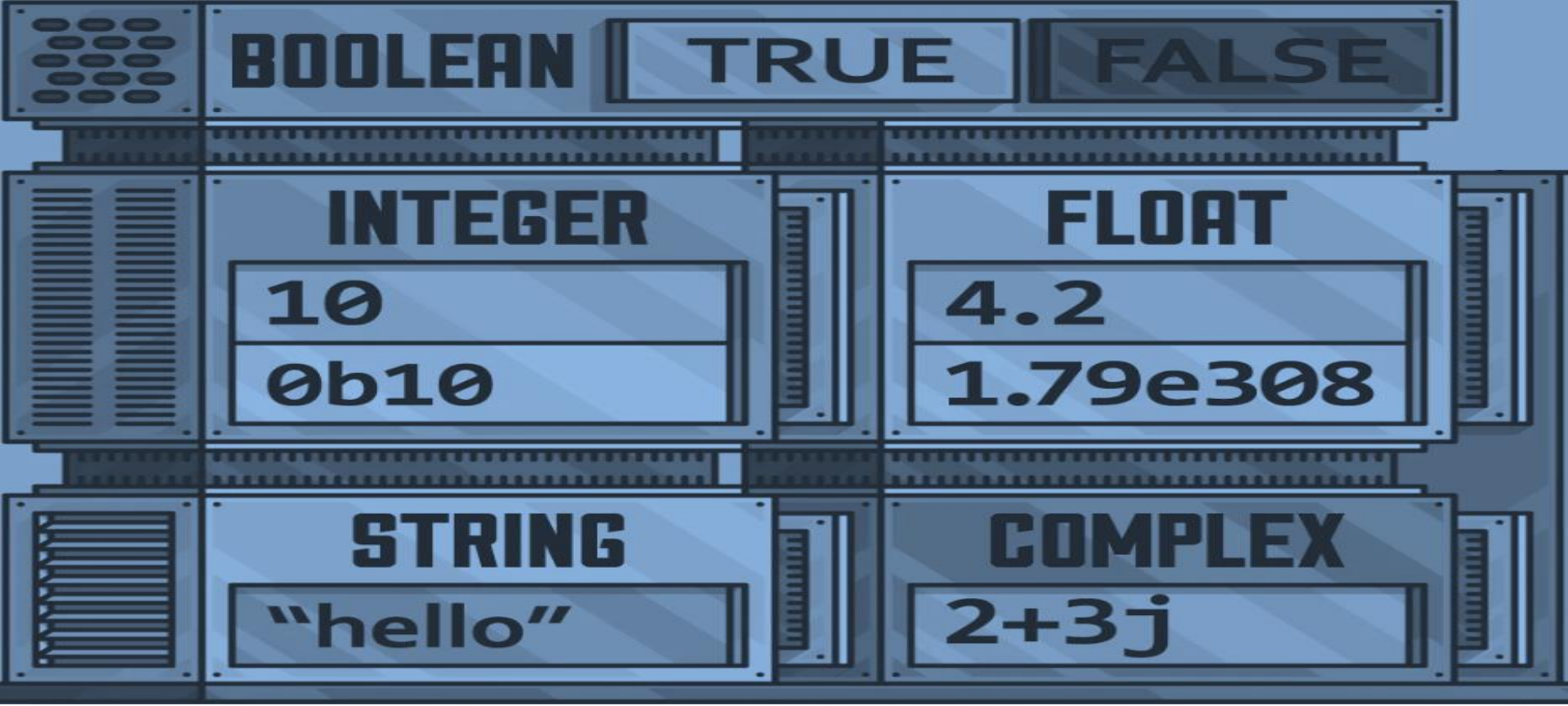

Java Programming

CPT111 – Lab 2

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CPT111 Java Programming

Lab 2

Data Types, Math, Client Input, Debugging

Lab Exercise #2.1

- In Lab Exercise #2.1, TA will lead you to **test**, **debug**, and **solve** a Math problem

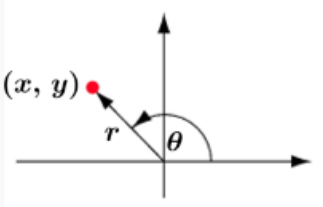
QUIZ

Lab Exercise #2.1 Polar Coordinates

Receive a grade

Opens: Thursday, 19 September 2024, 09:00
Closes: Wednesday, 25 September 2024, 23:59

Write a Java program that converts Cartesian coordinates to Polar coordinates.
Your program should take two real input numbers x and y, and print the Polar coordinates r and θ according to the formulas below.



$$r = \sqrt{x^2 + y^2}$$
$$\theta = \arctan\left(\frac{y}{x}\right)$$

Note that you should use the Java method `Math.atan2(y, x)`, which can compute the arctangent value of y/x in all quadrants.

Test case 1:
Input:
3.0
4.0
Output:
r = 5.0
theta = 0.9272952180016122

Test case 2:
Input:
1.0
1.0
Output:
r = 1.4142135623730951
theta = 0.7853981633974483

Lab Exercise #2.1 – Testing

- Two test cases are **given** in Lab Exercise #2.1
  - include them in your code
  - however, they are not enough to show that your code is correct...

- **Create** a few more test cases!
  - what would be better test cases that can be different enough from the given test cases?

```
Test case 1:
Input:
3.0
4.0
Output:
r = 5.0
theta = 0.9272952180016122

Test case 2:
Input:
1.0
1.0
Output:
r = 1.4142135623730951
theta = 0.7853981633974483
```

Lab Exercise #2.1 – Debugging

- Use the following **buggy** code that uses (incorrect) Math.atan function
  - you may also use **other** code that contains mistakes

```java
import java.util.Scanner;

public class PolarCoordinates {
    public static void main(String[] args) {
        Scanner kb = new Scanner(System.in);
        double x = Double.parseDouble(kb.nextLine());
        double y = Double.parseDouble(kb.nextLine());

        double r = Math.sqrt(x*x + y*y);
        double theta = Math.atan(y/x);

        System.out.println("r = " + r);
        System.out.println("theta = " + theta);
    }
}
```

simply use these three lines to get input into your program

always use nextLine() to get a string input, **then** convert to integer or double

- Use **debugger** and **visualizer** to show the lines that contain mistakes

Submitting Buggy Codes

- Codes that contain bug would **not** pass Learning Mall submission, for example:

	Input	Expected	Got	
✓	3.0 4.0	r = 5.0 theta = 0.9272952180016122	r = 5.0 theta = 0.9272952180016122	✓
✓	1.0 1.0	r = 1.4142135623730951 theta = 0.7853981633974483	r = 1.4142135623730951 theta = 0.7853981633974483	✓
✗	-1.0 1.0	r = 1.4142135623730951 theta = 2.356194490192345	r = 1.4142135623730951 theta = -0.7853981633974483	✗
✗	-1.0 -1.0	r = 1.4142135623730951 theta = -2.356194490192345	r = 1.4142135623730951 theta = 0.7853981633974483	✗
✓	1.0 -1.0	r = 1.4142135623730951 theta = -0.7853981633974483	r = 1.4142135623730951 theta = -0.7853981633974483	✓
✗	0.0 0.0	r = 0.0 theta = 0.0	r = 0.0 theta = NaN	✗

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

the code will *not* be able to pass these hidden test cases…

if it fails one or more hidden tests, then **your program has a bug !**

test and debug your code **with your own test cases** in your computer, until it passes enough test cases

for example, test with the input 0.0 and 0.0

```
run:
0.0
0.0
r = 0.0
theta = 0.0
```

Lab Exercise #2.1 – Solving

- The correct code should use Math.atan2 function (instead of Math.atan)
  - and just uses one Scanner object

```java

import java.util.Scanner;

public class PolarCoordinates {
    public static void main(String[] args) {
        Scanner kb = new Scanner(System.in);
        double x = Double.parseDouble(kb.nextLine());
        double y = Double.parseDouble(kb.nextLine());

        double r = Math.sqrt(x*x + y*y);
        double theta = Math.atan2(y, x);

        System.out.println("r = " + r);
        System.out.println("theta = " + theta);

        System.out.println("expected r = " + 0.0);
        System.out.println("expected theta = " + 0.0);
    }
}
```

- The TA will explain the solution
  - alternatively, you may watch the hint videos

MEDIASITE CONTENT
Lab Exercise #2.1 Hint Video 1

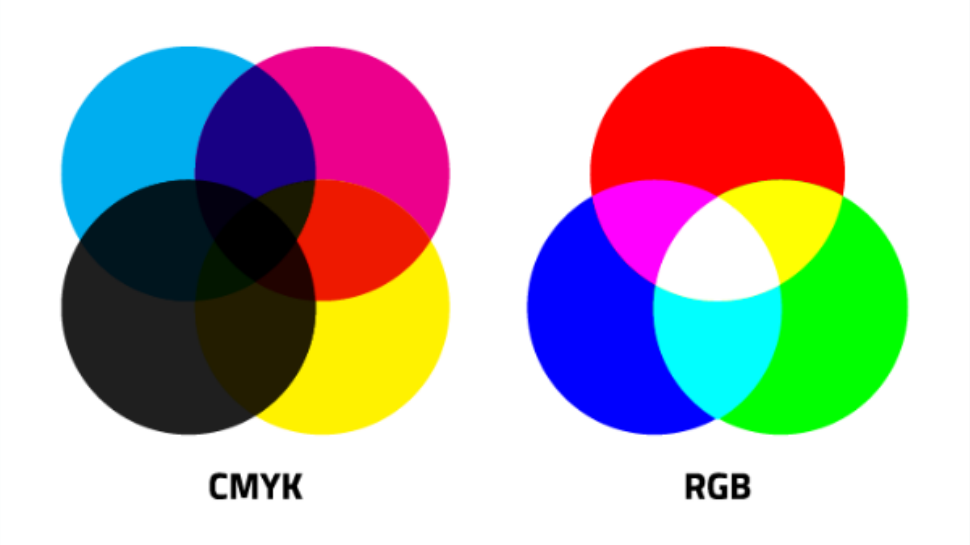
MEDIASITE CONTENT
Lab Exercise #2.1 Hint Video 2

Lab 2 Challenge – CMYK to RGB

- LM Quiz:

QUIZ
Lab 2 Challenge - CMYK to RGB

There are several different formats used to represent color.
You may have known the *RGB format* that specifies the level of red (R), green (G), and blue (B) on an *integer* scale from 0 to 255.
There is another format called the *CMYK format* that specifies the level of cyan (C), magenta (M), yellow (Y), and black (K) on a *real* scale from 0.0 to 1.0.



Lab 2 Challenge – CMYK to RGB

Write a Java program that converts from CMYK format to RGB format using the following formulas:

white = 1 − black

red = 255 × white × (1 − cyan)

green = 255 × white × (1 − magenta)

blue = 255 × white × (1 − yellow)

Your program must take four **`double`** input numbers cyan, magenta, yellow, and black;
compute the corresponding RGB values, each *rounded to the nearest integer*;
and print the RGB values as in the test cases below:

- There are 4 tasks to complete
  - each task is worth 25 marks
  - marked by one of your lab TAs

Lab 2 Challenge – Task 1: Create Test Cases (25 marks)

- You are provided with 2 given test cases

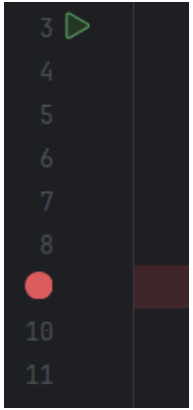
```
Test case 1:
Input:
0.0
0.0
1.0
0.0
Output:
red = 255
green = 255
blue = 0
```

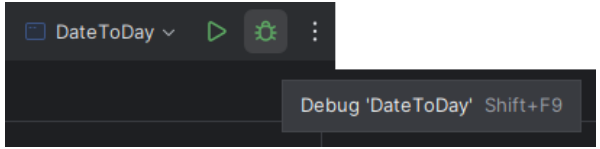
```
Test case 2:
Input:
0.0
1.0
0.75
0.50
Output:
red = 128
green = 0
blue = 32
```

- Create **at least two more** test cases, and tell your TA why you create them
  - you need to give **good reasoning** to earn a full mark in this task

Lab 2 Challenge – Task 2: Debug Your Code (25 marks)

- Write a code that contains one or more bugs
- Run a **debugger**, such as the one used in Lecture 2

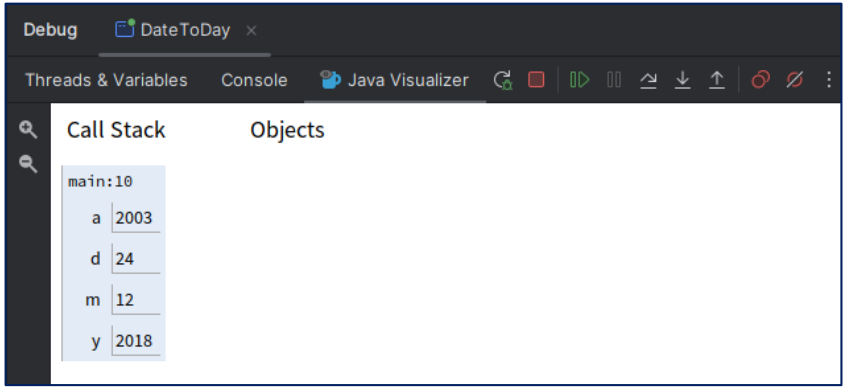




- Explain to TA **why** you suspect that there is a bug in that line
  - for example, the value in variable **`white`** does not match the correct value that you have computed manually

Lab 2 Challenge – Task 3: Visualize Your Code (25 marks)

- Run a **visualizer**, such as the one used in Lecture 2
  - you may use it as you debug your code



- Explain to the TA the **meaning** of the generated picture

Lab 2 Challenge – Task 4: Solve the Problem (25 marks)

- Solve the CMYK to RGB problem
- Show to the TA that you have passed the LM Quiz

	Input	Expected	Got	
✔	0.0 0.0 1.0 0.0	red = 255 green = 255 blue = 0	red = 255 green = 255 blue = 0	✔
✔	0.0 1.0 0.75 0.50	red = 128 green = 0 blue = 32	red = 128 green = 0 blue = 32	✔

Passed all tests! ✔

Correct

Marks for this submission: 100.00/100.00.

there are additional hidden tests

- Briefly explain to the TA you solution

Thank you for your effort!

- If you finish early, feel free to use the remaining time to work on other online exercises
  - use this opportunity to practice your new skills in testing, debugging, and visualizing your code

- You may leave the lab once you have complete the Lab 2 Challenge and it has been ASSESSED by a TA!
  - Please note that the Lab 2 Challenge is graded and contributes to your CW2 marks